DFCFN: Dual-Stage Feature Correction Fusion Network for Hyperspectral Pansharpening

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***Abstract*— Hyperspectral (HS) pansharpening aims to fuse high-spatial-resolution panchromatic (PAN) images with low-spatial-resolution HS (LRHS) images to generate high-spatial-resolution HS (HRHS) images. Due to the lack of consideration for the modal feature difference between PAN and LRHS images, most deep learning-based methods suffer from spectral and spatial distortions in the fusion results. In addition, most methods use upsampled LRHS images as network input, resulting in spectral distortion. To address these issues, we propose a dual-stage feature correction fusion network (DFCFN) that achieves an accurate fusion of PAN and LRHS images by constructing two fusion subnetworks: a feature correction compensation fusion network (FCCFN) and a multiscale spectral correction fusion network (MSCFN). Based on the lattice filter structure, FCCFN is designed to obtain the initial fusion result by mutually correcting and supplementing the modal features from PAN and LRHS images. To suppress spectral distortion and obtain fine HRHS results, MSCFN based on 2-D discrete wavelet transform (2D-DWT) is constructed to gradually correct the spectral features of the initial fusion result by designing a conditional entropy transformer (CE-Transformer). Extensive experiments on three widely used simulated datasets and one real dataset demonstrate that the proposed DFCFN achieves significant improvements in both spatial and spectral quality metrics over other state-of-the-art (SOTA) methods. Specifically, the proposed method improves the spectral angle mapping (SAM) metric by 6.4%, 6.2%, and 5.3% compared to the second-best comparison approach on the Pavia Center, Botswana, and Chikusei datasets, respectively. The codes are made available at: https://github.com/EchoPhD/DFCFN.**

***Index Terms*— 2-D discrete wavelet transform (2D-DWT), conditional entropy transformer (CE-Transformer), hyperspectral (HS) pansharpening, lattice filter.**

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I. Introduction

REMOTE sensing images with high spatial and spectral resolution have been widely used in various fields [1], including military surveillance [2], environmental monitoring [3], object identification [4], and classification [5]. However, due to the physical limitations of satellite sensors, hyperspectral (HS) images provide rich spectral information but have low spatial resolution. By contrast, panchromatic (PAN) images provide abundant spatial information but have lower spectral resolution. Therefore, researchers have proposed pansharpening methods that fuse spatial information from PAN images with spectral information from HS images to obtain high-spatial-resolution HS (HRHS) images. Although significant progress has been made in the pansharpening task, spectral and spatial distortions still exist in the fusion results. At present, HS pansharpening methods fall into two categories: traditional methods and deep learning (DL)-based methods.

Traditional pansharpening methods can be classified into four categories [6]: component substitution (CS)-based methods [7], [8], multiresolution analysis (MRA)-based methods [9], [10], Bayesian-based methods [11], [12], and model-based methods [13], [14], [15], [16]. Traditional methods can improve the spatial resolution of HS images. However, they often suffer from spatial and spectral distortions due to improper prior knowledge modeling [17], unknown sensor characteristics, and manually constructed features from dictionaries [18] that do not match actual features. In contrast, due to the powerful feature extraction capabilities of convolutional neural networks (CNNs), DL-based pansharpening methods have emerged [19], [20], [21], [22]. These fall into two categories according to network architectures, as shown in Fig. 1. One type adopts the architecture shown in Fig. 1(a), extracting PAN and upsampled low-spatial-resolution HS (LRHS) image features separately and combining them to obtain the final fusion results. The other approach adopts the architecture shown in Fig. 1(b), directly combining PAN images with upsampled LRHS images and sending them to a fusion network to fuse the two image features. The above two methods have two issues. On the one hand, they insufficiently consider the modal differences between spatial and spectral features, leading to spectral and spatial distortions in the fused images. On the other hand, due to the limited receptive field of CNNs, they cannot efficiently extract the global spatial and spectral information from the PAN and LRHS images.

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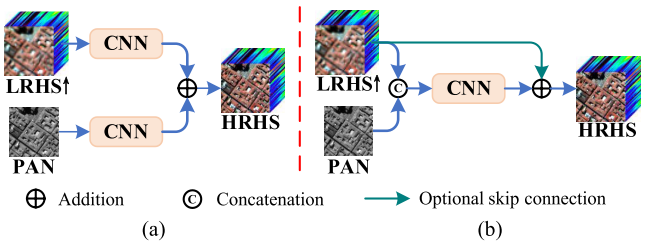


Fig. 1. Traditional CNN-based HS pansharpening network architectures. LRHS↑ represents the upsampled LRHS images. (a) Feature-based pansharpening. (b) Image-based pansharpening.

Nowadays, the transformer architecture has been successfully applied in the field of computer vision due to its ability to learn global dependencies of image features. Some researchers have also attempted to use transformer architecture to improve pansharpening performance by establishing the relationship between two modal features [23], [24]. However, compared to CNN, the transformer based on the self-attention mechanism has limited capability in extracting local features. Therefore, in the HS pansharpening task, it is challenging to use only the transformer for the fusion of PAN and LRHS images.

For HS pansharpening, the initial step typically involves upsampling the LRHS images, as shown in Fig. 1. However, the simple interpolation operation used for upsampling LRHS images may directly increase the degree of spectral distortion in the images. To illustrate this situation, we present the spectral changes of ground truth (GT) and LRHS images at the same locations, as shown in Fig. 2. GT and LRHS images are downsampled and upsampled with a scale factor of 4, respectively. Fig. 2(a) shows that LRHS and downsampled GT images have similar spectral information. However, there are significant spectral differences between the GT and upsampled LRHS images, as shown in Fig. 2(b), indicating that the upsampling operation causes spectral distortion. This verifies that using upsampled LRHS images as network input for fusion will inevitably introduce spectral distortion. To reduce the negative impact caused by the upsampling operation, He et al. [19] introduced implicit neural representations and constructed dynamic spatial–spectral encoding and spatial–spectral query mappings, achieving better performance in arbitrary-resolution HS pansharpening. Dong et al. [25] constructed a progressive perceptual model-driven scale-matched recovery network and a cross-scale model-driven upsampling network by utilizing diffusion models to achieve better fusion results. In this work, the features of the original LRHS images are taken as guiding features to supplement and correct the spectral information to effectively suppress the spectral distortion.

Based on the above analysis, we propose a dual-stage feature correction fusion network (DFCFN) that gradually corrects the two modal features to fuse PAN and LRHS images. This network fully considers the interactive fusion of spatial and spectral features and utilizes the mapping relationship between LRHS images and HRHS images to perform spatial and spectral corrections on the fusion results. Specifically, in the first stage, based on the lattice filter structure from the field of signal processing, which can correct system characteristics and reduce the impact of signal interference, we design a lattice fusion architecture to reduce the differences between the two modal features. Using this fusion structure, we then construct a feature correction compensation fusion network (FCCFN) to achieve intermodal interactive fusion and obtain the initial fusion result. In the second stage, to suppress spectral distortion in the initial fusion results, we design a multiscale spectral correction fusion network (MSCFN) based on 2-D discrete wavelet transform (2D-DWT) and a conditional entropy transformer (CE-Transformer) under the guidance of LRHS image features. The contributions of this article are as follows.

1) We propose a DFCFN consisting of two subnetworks, FCCFN and MSCFN, to progressively correct and fuse the modal features from PAN and LRHS images.
2) To reduce the difference between the two modal features, a lattice fusion architecture is designed. On this basis, FCCFN is constructed to achieve the initial fusion of PAN and LRHS images by correcting and supplementing the two image features with each other.
3) To suppress spectral distortion in the initial fusion result, we design the MSCFN based on 2D-DWT and CE-Transformer to generate the final fusion image. At each-level DWT layer in MSCFN, a CE-Transformer module is constructed to utilize the LRHS image features to correct features of different scales from the initial fusion result.
4) To validate the effectiveness of the proposed network, we conduct a large number of experiments on multiple datasets. The experimental results demonstrate that this method has superior performance compared with other state-of-the-art (SOTA) methods.

II. Related Work

A. Traditional HS Pansharpening Methods

In traditional HS image fusion methods, CS-based methods mainly include the principal component analysis (PCA) method [8] and intensity–hue–saturation (IHS) method [26]. The idea of these methods is to replace the spatial components of HS images with those of PAN images to improve the spatial resolution of the fused HRHS images. However, due to the sensitivity to factors such as noise and linear assumptions, these methods often suffer from severe spectral distortions. MRA-based methods inject spatial information obtained from multiscale decomposition components of PAN images into the HS images. The classic MRA-based methods used in these methods mainly include smooth filter intensity modulation (SFIM) [10] and high-pass modulation MTF-GLP (MTF-GLP-HPM) [27]. Compared with CS-based methods, MRA-based methods better preserve spectral information but are more sensitive to registration errors, resulting in distortions of the spatial information. Combining the advantages of CS and MRA, some researchers have proposed hybrid algorithms, such as guided filter PCA (GFPCA) [28]. Bayesian-based methods use the Bayesian inference framework to solve fusion problems by constructing prior regularization terms such as convex regularization (Hysure) [12] and Bayesian

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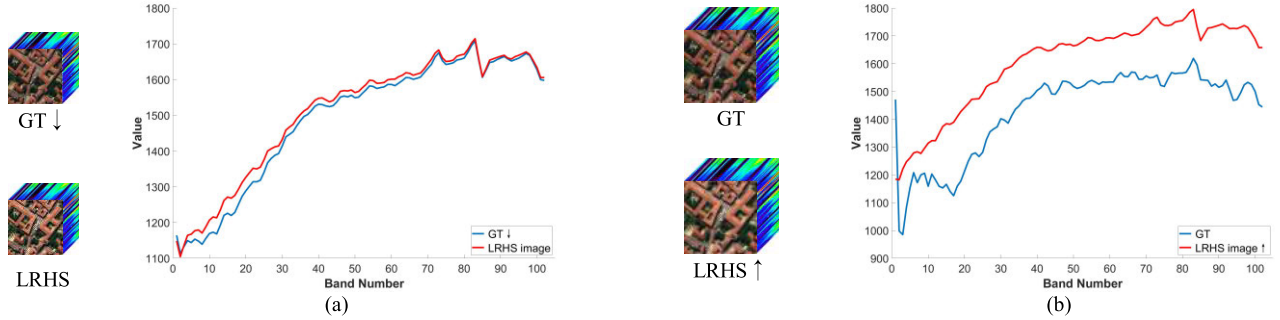


Fig. 2. Relationship between the spectral curves of GT and LRHS images. GT↓ denotes the GT images downsampled by a factor of 4, and LRHS↑ represents the LRHS images upsampled by a factor of 4. (a) Pavia Center (10, 20). (b) Pavia Center (40, 80).

sparse-promoting (BSP) Gaussian prior [29]. Model-based methods [14], [30], [31] use prior regularizations based on spectral unmixing for fusion problems. Traditional methods are data-independent and characterized by high efficiency. However, they often suffer from spatial and spectral distortions due to unknown sensor characteristics and manually constructed features from dictionaries that do not match actual features.

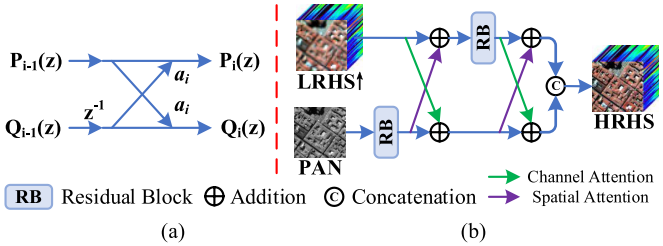


Fig. 3. LFS. (a) Lattice filter unit. (b) Lattice fusion architecture.

B. DL-Based HS Pansharpening Methods

Early DL-based HS pansharpening methods mainly adopted CNNs for feature extraction. Li et al. [32] proposed a cross-scale nonlocal attention network to obtain reconstructed HRHS images by learning the spatial and spectral correlations between the HS and PAN image features. Sun et al. [33] constructed the dual-branch interactive network AIDB-Net and proposed a spectral–spatial interactive attention mechanism for achieving global cross-modal feature fusion. Zhuo et al. [34] proposed the Hyper-DSNet method, which preserves spatial details and spectral fidelity by constructing a deep-shallow fusion structure with multidetail extraction and spectral attention. To address the spectral distortion caused by upsampling LRHS images, Zheng et al. [35] proposed an HS pansharpening method based on deep image prior and attention structures, replacing the LRHS images upsampling process with a deep image prior network and concatenating multiple parallel channels and spatial attention residual blocks (RBs) to learn fusion features. Bandara et al. [36] proposed an HS pansharpening method based on an improved deep image prior and residual reconstruction, which improves the quality of fusion results by enhancing the quality of upsampled HS images and constructing a network called HyperKite that focuses on learning high-level features. Although these CNN-based methods have better performance than traditional methods, the local feature learning ability of CNNs hinders the exploration and utilization of long-range correlations of image features in networks.

Recently, the transformer-based HS pansharpening methods have been proposed because of the powerful global feature extraction capability of transformer. Peng et al. [37] proposed a double U-Net structure that fuses spatial–spectral information from PAN images and HS images through a spatial–spectral integration structure. Zhou et al. [38] constructed an upsampling spectral-preserving network to mitigate spectral distortion and achieved multiscale feature enhancement and fusion by incorporating self-attention and cross-attention modules. Shang et al. [24] modeled the pansharpening task as an optimization model and utilized the global information representation ability of transformer for optimization and solution. Bandara and Patel [39] constructed a HyperTransformer for pansharpening to learn the cross-feature space dependencies and long-range features of PAN and LRHS images. Although transformer-based pansharpening models have demonstrated impressive performance, they lack local feature extraction capability. Therefore, there remains significant potential for further enhancing transformer-based model performance.

C. Lattice Fusion Structure

In signal processing, the basic unit of the standard lattice filter [40] is a locally correlated module structure called an adaptive equalizer, as shown in Fig. 3(a). This structure adjusts the filter coefficients based on the characteristics of the input signals, helping to correct the time-varying characteristics of the system and reduce interference between signals. The operation of a basic unit in the standard lattice filter can be defined as follows:

$$P_i(z) = P_{i-1}(z) + a_i z^{-1} Q_{i-1}(z) \tag{1}$$

$$Q_i(z) = a_i P_{i-1}(z) + z^{-1} Q_{i-1}(z) \tag{2}$$

where $P_i(z)$ and $Q_i(z)$ represent the outputs of the basic unit, $P_{i-1}(z)$ and $Q_{i-1}(z)$ represent the inputs of the basic unit, z^{-1} corresponds to the delay of one unit of the signal in the time domain, and a_i represents the filtering coefficient of two intersecting paths.

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Inspired by the lattice filter, we design a lattice fusion structure (LFS) as shown in Fig. 3(b) to the difference between the two modal features. LFS can reduce the interference of differences between the two features on the fusion results by correcting and supplementing the modal features. To better correct the two features, LFS uses two lattice filter units, each of which utilizes an RB for extracting and integrating these two features. The coefficients on the two intersecting paths are adaptively obtained by performing spatial attention and channel attention operations on the two image features.

III. Proposed Method

In this section, we present our proposed DFCFN, as shown in Fig. 4. DFCFN has two fusion stages: FCCFN and MSCFN. In the first stage, FCCFN reduces the differences between the two model features to produce the initial fusion result of the PAN and upsampled LRHS images. In the second stage, to suppress the spectral distortion in the initial fusion results, MSCFN is designed. First, multilevel 2D-DWTs are used to decompose the initial fusion result into high- and low-frequency components, and then, the spectral features of the LRHS image are used as guiding features to reconstruct the fine fusion result. In MSCFN, a CE-Transformer in each-level DWT layer is constructed to correct the low-frequency components of the initial fusion result using the reconstructed IDWT features or the LRHS images features. We provide a detailed introduction to the FCCFN, MSCFN, and CE-Transformer structures as follows.

A. Feature Correction Compensation Fusion Network

CNN-based pansharpening methods generally adopt the network architectures shown in Fig. 1(a) and (b); neither architecture interactively adjusts the modal differences between PAN and upsampled LRHS image features. However, these differences are one of the key factors leading to spectral and spatial distortions in the fusion results. Therefore, we build FCCFN based on LFS to reduce these differences while supplementing and fusing the image features.

Based on the lattice fusion architecture in Fig. 3(b), FCCFN has the design shown in Fig. 4(a) and takes PAN and upsampled LRHS images as inputs. Considering the rich spectral information of upsampled LRHS images, the cross coefficients between the two paths from bottom to top are obtained using channel attention and are used to supplement and correct the spectral features of the PAN images. Conversely, PAN images have rich spatial information, and the cross coefficients between two paths from top to bottom are obtained using spatial attention and are used to supplement and correct the spatial features of the upsampled LRHS images. Therefore, the first lattice unit of LFS is expressed as

$$F_{\text{pan}}^{1} = \text{RB}(I_{\text{PAN}}) + \text{CA}\left(I_{\text{lrhs}\uparrow}\right) \tag{3}$$

$$F_{\text{hs}}^{1} = I_{\text{lrhs}\uparrow} + \text{SA}(\text{RB}(I_{\text{PAN}})) \tag{4}$$

where $\text{SA}(\cdot)$ and $\text{CA}(\cdot)$ denote spatial attention and channel attention operations, respectively. Here, we adopt the spatial attention and channel attention structures from the CBAM network proposed by Woo et al. [41]. $\text{RB}(\cdot)$ represents an RB structure, consisting of two convolution layers with a kernel size of 3×3 and a ReLU activation function. F_{pan}^{1} and F_{hs}^{1} represent the corrected spatial and spectral features in the first lattice unit of FCCFN, respectively. Similarly, the second lattice unit of LFS is defined as

$$F_{\text{hs}}^{2} = \text{RB}\left(F_{\text{hs}}^{1}\right) + \text{SA}\left(F_{\text{pan}}^{1}\right) \tag{5}$$

$$F_{\text{pan}}^{2} = F_{\text{pan}}^{1} + \text{CA}\left(F_{\text{hs}}^{1}\right) \tag{6}$$

where F_{hs}^{2} and F_{pan}^{2} represent the corrected spectral and spatial features obtained by the second lattice unit of FCCFN, respectively.

At the end of FCCFN, the outputs of the two paths are concatenated and fed into a convolutional layer to obtain an initial fusion result. The fusion operation of the first stage can be represented as

$$I_{\text{one}} = \text{Conv}_{1\times1}\left(\text{Cat}\left(F_{\text{pan}}^{2}, F_{\text{hs}}^{2}\right)\right) \tag{7}$$

where $\text{Conv}_{1\times1}(\cdot)$ represents a convolution layer with a kernel size of 1×1, $\text{Cat}(\cdot)$ represents the concatenation operation, and I_{one} represents the initial fusion result output by the first stage.

B. Multiscale Spectral Correction Fusion Network

1) Network Structure: The first fusion stage achieves correction and fusion of two modal features in high-resolution space, but the spectral distortion introduced by upsampling LRHS images is not improved. Therefore, the initial fusion result I_{one} still has spectral and spatial distortions. We verify this in Section IV-D1. In the second fusion stage, considering the advantages of wavelet transform in the context of multiscale transformation [42], [43], [44], Zhang et al. [43] developed a wavelet-based feature enhancement and fusion network for the task of pansharpening, which capitalizes the inherent spatial-frequency interaction and frequency division capabilities within the wavelet transform. Motivated by the DePF structure [44], we design an MSCFN based on multilevel 2D-DWTs and CE-Transformer, as shown in Fig. 4(b). Considering that the LRHS image I_{lrhs} retains the original spectral information, MSCFN takes the features of I_{lrhs} as guiding features to gradually supplement and correct the spectral information of the initial fusion result I_{one} from downscaling to the original scale space to obtain fine HRHS results. In MSCFN, a CE-Transformer is designed by establishing the correlation between the features from the LRHS image I_{lrhs} and features at different scales of initial fusion result I_{one} to correct the spectral features of I_{one}. Because the spatial size of the initial fusion result is four times that of the LRHS image, MSCFN uses three-level 2D-DWTs.

The specific operation of MSCFN is as follows. First, three-level 2D-DWTs are performed on I_{one} to obtain four components $[ll^{i}, lh^{i}, hl^{i}, hh^{i}](i = 1, 2, 3)$ that represent the low-frequency component as well as the high-frequency components of I_{one} in the vertical, horizontal, and diagonal directions. Among them, the low-frequency component mainly contains the spectral information of I_{one}, while the other three components mainly contain spatial texture information.

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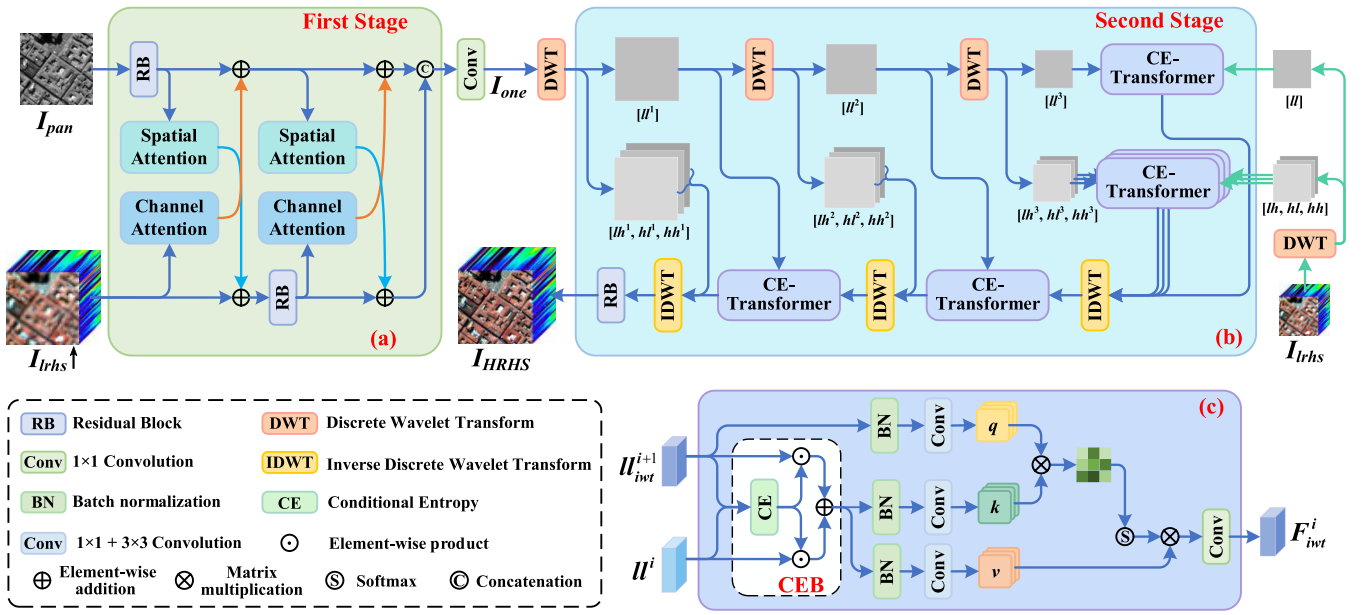


Fig. 4. Overall architecture of DFCFN. FCCFN denotes feature correction compensation fusion network, MSCFN denotes multiscale spectral correction fusion network, CEB denotes conditional entropy block, and CE-Transformer represents conditional entropy transformer.

Meanwhile, one-level 2D-DWT is performed on the LRHS image I_{lrhs} to obtain four components $[ll, lh, hl, hh]$. The above operations can be expressed as

$$[ll^1, lh^1, hl^1, hh^1] = \text{DWT}(I_{\text{one}}) \tag{8}$$

$$[ll^{i+1}, lh^{i+1}, hl^{i+1}, hh^{i+1}] = \text{DWT}(ll^i), \quad i = [1, 2] \tag{9}$$

$$[ll, lh, hl, hh] = \text{DWT}(I_{\text{lrhs}}) \tag{10}$$

where $\text{DWT}(\cdot)$ denotes the 2D-DWT operation.

Then, feature fusion operations are performed on low-frequency components at three scales. At the third scale, because the LRHS image mainly contains spectral information, four components $D_h^3 \in [ll^3, lh^3, hl^3, hh^3]$ of the third-level 2D-DWT are fused with four components $D_l \in [ll, lh, hl, hh]$ of the LRHS image through four CE-Transformer modules. The 2-D inverse discrete wavelet transform (2D-IDWT) is applied to the four fusion components to obtain the fused components at the current scale. These operations can be expressed as

$$F_i^3 = \text{CETF}_i\left(D_h^3, D_l\right), \quad i = [1, 2, 3, 4] \tag{11}$$

$$ll_{\text{iwt}}^3 = \text{IDWT}\left(F_1^3, F_2^3, F_3^3, F_4^3\right) \tag{12}$$

where $\text{CETF}(\cdot)$ denotes the operation of the CE-Transformer, F_i^3 represents the four fusion components at the third scale, $\text{IDWT}(\cdot)$ denotes the 2D-IDWT operation, and ll_{iwt}^3 represents the result of 2D-IDWT at the third scale.

At the second and first scales, the results ll_{iwt}^3 and ll_{iwt}^2 of 2D-IDWTs are fused with low-frequency components ll^2 and ll^1 of 2D-DWTs to obtain the fused low-frequency components. These fused components, along with the other three components from 2D-DWTs, participate in 2D-IDWTs to reconstruct fusion results ll_{iwt}^i at each scale. Finally, two sequentially connected RBs are utilized to refine features and generate the final fusion result. The above process can be expressed as

$$ll_{\text{iwt}}^i = \text{IDWT}\left(\text{CETF}\left(ll^i, ll_{\text{iwt}}^{i+1}\right), lh^i, hl^i, hh^i\right) \tag{13}$$

$$I_{\text{HRHS}} = \text{RB}\left(\text{RB}\left(ll_{\text{iwt}}^1\right)\right) \tag{14}$$

where $ll_{\text{iwt}}^i (i = 1, 2)$ denotes the results of 2D-IDWTs at the first and second scales and I_{HRHS} denotes the final fusion image.

2) CE-Transformer: To utilize spectral features from the LRHS image to correct spectral information in the initial fusion result, we design a CE-Transformer, as shown in Fig. 4(c). Note that the CE-Transformer structure, constructed for the fusion of low-frequency features, is also applied to all four components of I_{lrhs} at the third scale to better integrate spectral information from I_{lrhs}. For clarity, we will provide a detailed explanation of the CE-Transformer structure using the fusion of low-frequency features as an example.

First, we design a CE block (CEB) using the idea of CE [45]. CEB handles the interaction and supplementation of two types of input information by evaluating the differences between the low-frequency components ll^i and the fused result ll_{iwt}^{i+1}. The low-frequency component ll^i of each scale obtained by 2D-DWT mainly contains spectral information, but it is not completely consistent with the spectral information of I_{lrhs}. Therefore, we introduce the idea of CE to correct and supplement the fusion features at each scale by evaluating the differences between them. Based on the theory of CE, the low-frequency components ll^i and the fused result ll_{iwt}^{i+1} can be considered two random variables. To describe one of these two variables, CE can be used to calculate the additional information required under the condition that one variable is known. Based on this idea, we obtain the weight coefficients for calculating additional information using the following formulas:

$$H\left(ll^i | ll_{\text{iwt}}^{i+1}\right) = -\sum P\left(ll^i, ll_{\text{iwt}}^{i+1}\right) * \log_2\left(P\left(ll^i | ll_{\text{iwt}}^{i+1}\right)\right) \tag{15}$$

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TABLE I
AVERAGE QUANTITATIVE PANSHARPENING RESULTS ON THE PAVIA CENTER, BOTSWANA, AND CHIKUSEI DATASETS

Dataset	Methods	PSNR(↑)	SAM(↓)	SSIM(↑)	SCC(↑)	RMSE×10^{-2}(↓)	ERGAS(↓)
Pavia center	SFIM [10]	28.2767	8.6505	0.8222	0.8965	4.1160	7.431
	MTF_GLP_HPM [27]	31.3672	8.8859	0.8911	0.9267	3.1100	5.9379
	GSA [7]	30.0220	8.4579	0.8480	0.9263	3.4710	5.9508
	CNMF [14]	32.0244	7.2364	0.8913	0.9462	2.7090	4.8195
	SSFCNN [50]	35.589	5.7787	0.9426	0.9761	1.8104	3.3362
	HyperPNN [51]	37.0370	5.2954	0.9520	0.9806	1.5656	2.9419
	DHP-DARN [35]	37.1592	5.2685	0.9545	0.9824	1.5299	2.8876
	DIP-Hyperkite [36]	36.9586	5.3630	0.9527	0.9809	1.5661	2.9363
	Hyper-DSNet [34]	37.1725	5.1541	0.9539	0.9822	1.5245	2.8929
	HyperRefiner [38]	37.7193	5.0461	0.9588	0.9856	1.4265	2.7748
	Ours	**38.5797**	**4.7225**	**0.9621**	**0.9870**	**1.3112**	**2.5485**
Botswana	SFIM [10]	34.1704	2.4259	0.9104	0.9557	4.2630	3.8398
	MTF_GLP_HPM [27]	36.9958	2.2292	0.9352	0.9637	3.1770	2.7787
	GSA [7]	36.3937	2.3272	0.9253	0.9631	3.1750	2.9275
	CNMF [14]	33.6006	2.3018	0.9193	0.9549	4.5080	3.8151
	SSFCNN [50]	42.3542	1.9164	0.9386	0.9818	1.3880	4.1864
	HyperPNN [51]	42.5861	1.9909	0.9522	0.9778	1.3955	1.8251
	DHP-DARN [35]	41.1629	1.9877	0.94533	0.9818	1.3589	2.5892
	DIP-Hyperkite [36]	42.4773	1.7728	0.9576	0.9825	1.2897	1.8724
	Hyper-DSNet [34]	43.7004	1.8418	0.9554	0.9809	1.2998	1.4724
	HyperRefiner [38]	44.2455	1.6751	0.9663	0.9864	1.1655	1.3997
	Ours	**44.9590**	**1.5767**	**0.9687**	**0.9881**	**1.0830**	**1.3166**
Chikusei	SFIM [10]	29.9977	6.8942	0.8438	0.7233	4.0140	25.7210
	MTF_GLP_HPM [27]	30.6859	7.7090	0.8458	0.7052	4.5440	32.0045
	GSA [7]	35.4329	4.2217	0.9095	0.9136	1.9210	8.0092
	CNMF [14]	35.6701	3.7933	0.8970	0.9013	1.9730	6.6018
	SSFCNN [50]	40.8220	2.4829	0.9656	0.9712	1.0188	4.3537
	HyperPNN [51]	40.6471	2.5668	0.9644	0.9699	1.0318	4.5882
	DHP-DARN [35]	41.0419	2.4217	0.9686	0.9743	0.9837	4.2614
	DIP-Hyperkite [36]	41.6503	2.3207	0.9702	0.9761	0.9246	4.0896
	Hyper-DSNet [34]	41.6990	2.3315	0.9705	0.9765	0.9197	4.0150
	HyperRefiner [38]	42.9958	2.1112	0.9777	0.9829	0.8046	3.5126
	Ours	**43.4760**	**1.9978**	**0.9792**	**0.9843**	**0.7680**	**3.3169**

$$H\left(ll^i_{\text{iwt}}|ll^i\right) = -\sum P\left(ll^i, ll^i_{\text{iwt}}\right) * \log_2\left(P\left(ll^i_{\text{iwt}}|ll^i_{\text{iwt}}\right)\right) \tag{16}$$

where $H(a|b)$ denotes the weight coefficients obtained using b as the known condition, which is used to calculate the supplementary information provided to a. $P(a, b)$ represents the joint probability distribution of a and b, $\log_2(\cdot)$ denotes logarithm with base 2, and $P(a|b)$ represents the conditional probability distribution of a given b. Afterward, the weighting operations are performed on the two input features to calculate the supplementary features and then integrate them. The process can be expressed as

$$F_1^{\text{dwt}} = ll^i * H\left(ll^i|ll^i_{\text{iwt}}\right) \tag{17}$$

$$F_2^{\text{dwt}} = ll^i_{\text{iwt}} * H\left(ll^i_{\text{iwt}}|ll^i\right) \tag{18}$$

$$F_{\text{CE}} = \text{CEB}\left(ll^i, ll^i_{\text{iwt}}\right) = F_1^{\text{dwt}} + F_2^{\text{dwt}} \tag{19}$$

where F_1^{dwt} and F_2^{dwt} represent the adjusted spectral features obtained by CE and F_{CE} represents the output of the CEB.

The integrated spectral features F_{CE} are then mapped into two vectors in the transformer: key (k) and value (v), and the low-frequency components ll^i of the initial fusion image are mapped into Query(q). q and k are used to establish the correlation matrix w_c between low-frequency features and supplemented spectral features. These operations can be represented as

$$v = \text{shape}(\text{Conv}_{3\times3}(\text{Conv}_{1\times1}(F_{\text{CE}})) \tag{20}$$

$$\text{k} = \text{shape}(\text{Conv}_{3\times3}(\text{Conv}_{1\times1}(F_{\text{CE}})) \tag{21}$$

$$q = \text{shape}(\text{Conv}_{3\times3}(\text{Conv}_{1\times1}(ll^i)) \tag{22}$$

$$w_c = \text{Softmax}(\text{MatMul}(q, k^T)) \tag{23}$$

where $\text{Conv}(\cdot)_{3\times3}$ represents the convolution operation with a kernel size of 3×3, shape($\cdot$) represents the reshape operation, MatMul($\cdot$) represents the matrix multiplication operation, and Softmax($\cdot$) represents the softmax operation.

Finally, the correlation matrix is used to weight v, with the corrected spectral features obtained through a linear operation defined by

$$\begin{aligned} F_{\text{iwt}}^{i-1} &= \text{CETF}\left(ll^i, ll^i_{\text{iwt}}\right) \\ &= \text{Conv}_{1\times1}(\text{shape}(\text{MatMul}(w_c, v))) \end{aligned} \tag{24}$$

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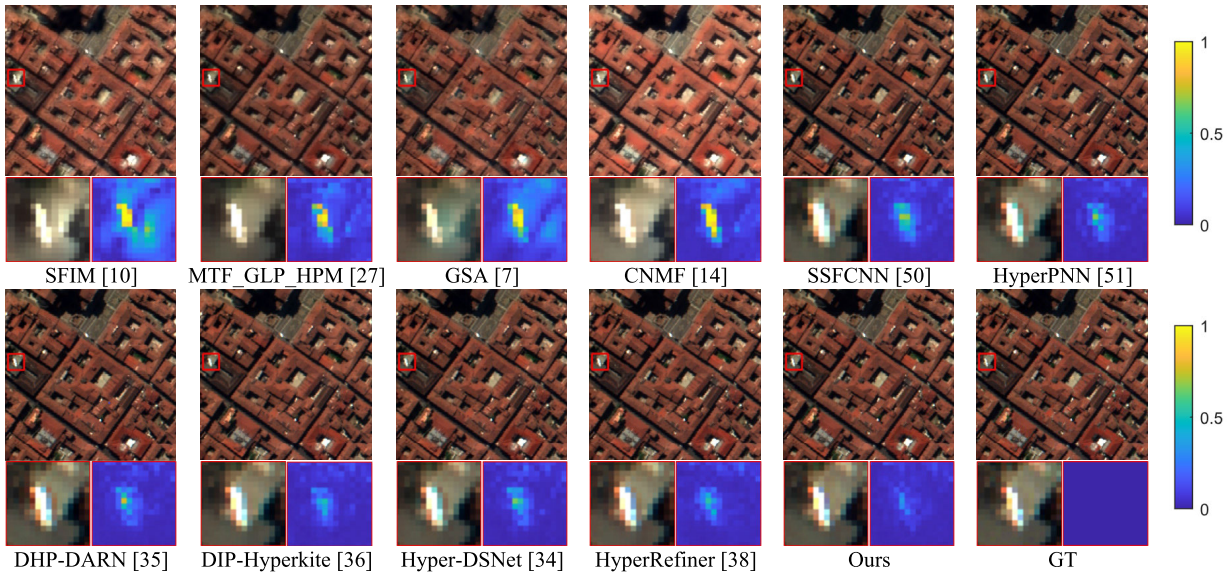


Fig. 5. Fusion results on the Pavia Center dataset.

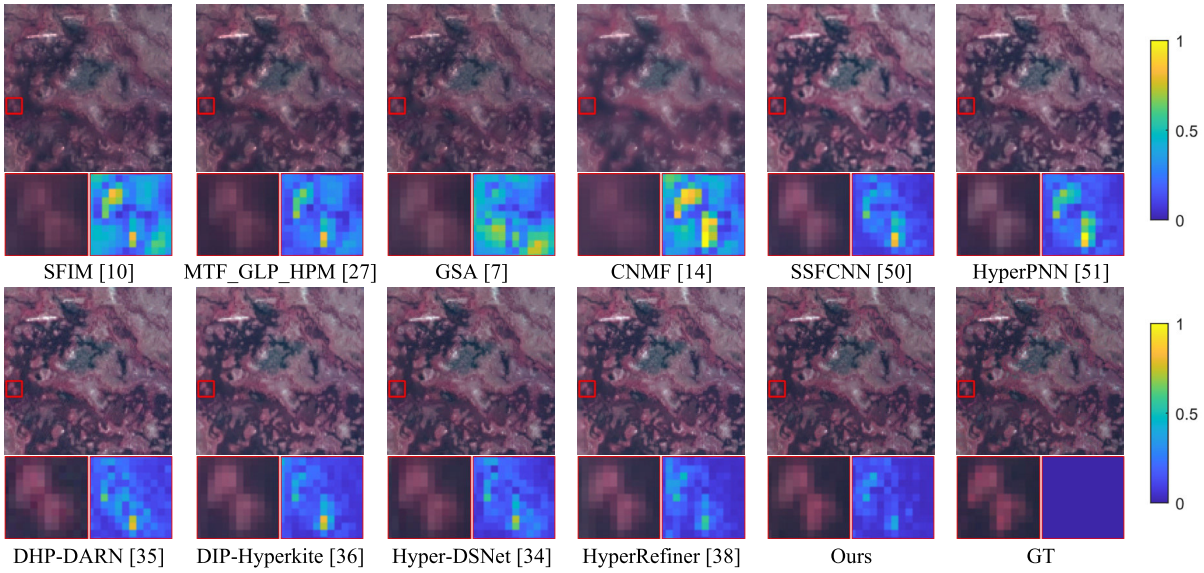


Fig. 6. Fusion results on the Botswana dataset.

where T represents the transpose operation and F_{iwt}^{i-1} represents the output of CE-Transformer at the $(i-1)$th scale.

C. Loss Function

L_1 loss is used to calculate the reconstruction loss because it has been demonstrated to perform better than the L_2 loss in pansharpening [34]. The reconstruction loss is defined as

$$L_{\text{ref}} = \frac{1}{CHW}\|H_{\text{GT}} - I_{\text{HRHS}}\|_1 \tag{25}$$

where $H_{\text{GT}} \in \mathbb{R}^{C\times H\times W}$ denotes the GT image; I_{HRHS} is the HRHS image predicted by the network; and (C, H, W) denotes the dimensions of H_{GT} in terms of the number of spectral channels, height, and width.

IV. Experiments

In this section, extensive experiments are conducted on three widely used available simulated datasets and one real dataset, and several SOTA methods are compared to demonstrate the effectiveness of the proposed DFCFN.

A. Datasets and Experimental Setup

To prove the effectiveness of our proposed DFCFN, we conducted extensive experiments on three publicly available and widely used simulated HS datasets: Pavia Center [46], Botswana [47], and Chikusei [48]. For network training, the datasets were constructed following Wald [49] and set according to Bandara [36]. In the Pavia Center, Botswana, and Chikusei datasets, the sizes of the GT images are $160 \times 160 \times 102$, $120 \times 120 \times 145$, and $256 \times 256 \times 128$, respectively, and the sizes of LRHS images are $40 \times 40 \times 102$, $30 \times 30 \times 145$, and $64 \times 64 \times 128$, respectively. When generating RGB images for the synthetic Pavia Center, Botswana, and Chikusei datasets, we employed the 10th, 10th, and 12th spectral bands as the blue channels; the 30th, 35th, and 20th spectral bands as the green channels; and the 60th, 61st, and 29th spectral

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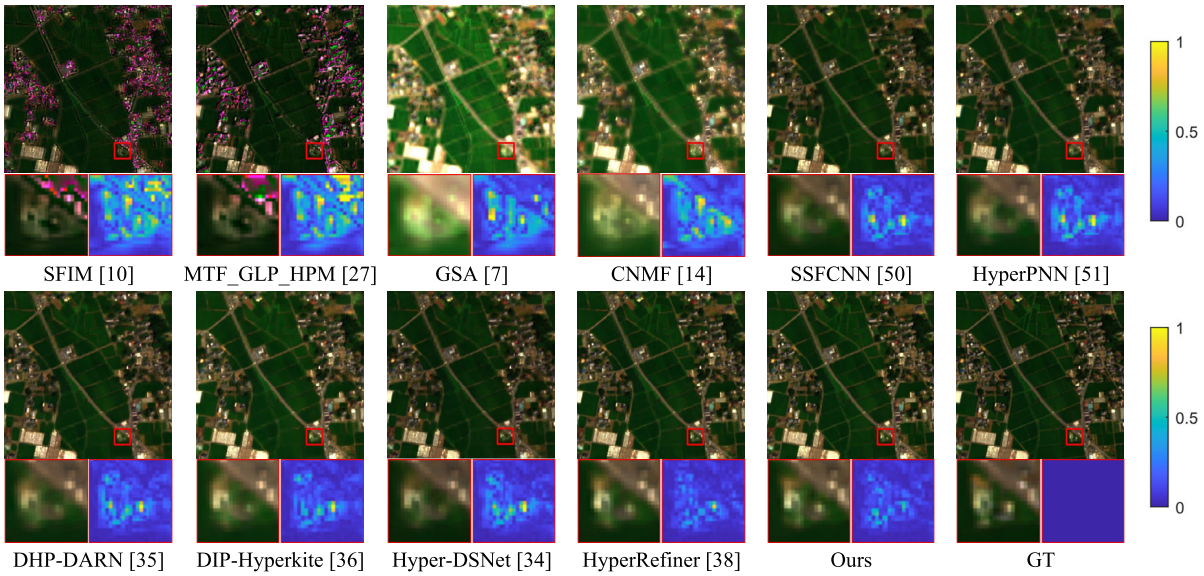


Fig. 7. Fusion results on the Chikusei dataset.

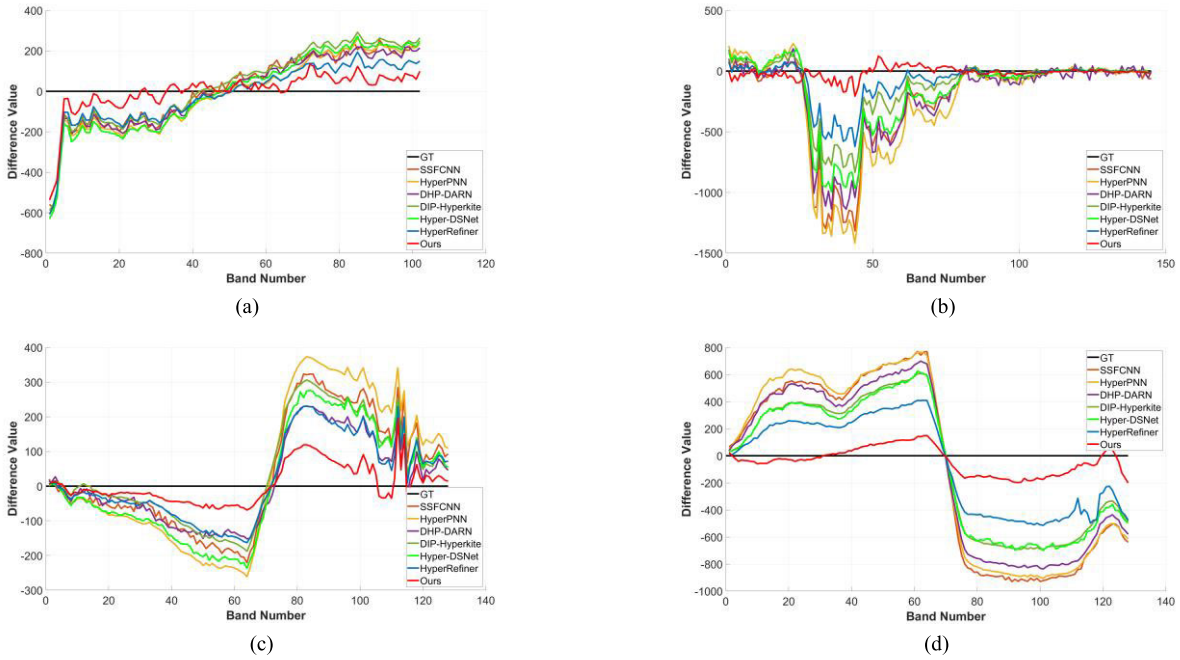


Fig. 8. Spectral differences at three locations between the GT and pansharpening results from Fig. 7. (a) Pavia Center (89, 130). (b) Botswana (60, 10). (c) Chikusei (87, 108). (d) Chikusei (171, 215).

bands as the red channels. To further validate the effectiveness of the proposed model, we conducted experiments on the real HS pansharpening dataset FR1 [34], which can be obtained from the following website (https://openremotesensing.net/HS-pansharpening-challenge/). The dataset contains precisely co-registered image pairs of PAN (2400 × 2400 pixels) and LRHS (400 × 400 × 69 pixels). In the real experiments, to match the 1:4 ratio of LRHS and PAN image sizes, we first downsampled the PAN images from a size of 2400 × 2400–1600 × 1600. Then, six pairs of images (LRHS image with a size of 80 × 80 × 69 and PAN image with a size of 320 × 320) were cropped from the original image for testing.

For fair comparisons, we retrained all DL-based methods using Python 3.9 and PyTorch 1.13 on the Ubuntu 20.04 system with NVIDIA RTX A6000. The initial learning rate, epoch, and batch size are set to 0.0001, 8000, and 4, respectively. Adam is selected as the optimizer, and the learning rate decays by 0.8 every 2000 epochs.

B. Comparison Methods and Quantitative Metrics

To verify the effectiveness of the proposed DFCFN, we compared it with other SOAT methods, including four traditional methods (CNMF [14], GSA [7], MTF_GLP_HPM [27], and SFIM [10]) and six

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TABLE II
AVERAGE QUANTITATIVE ON THE REAL FR1 DATASET

Methods	D_λ (↓)	D_s (↓)	QNR (↑)
SFIM [10]	0.0436	0.0572	0.9016
MTF_GLP_HPM [27]	0.1002	0.0914	0.8175
GSA [7]	0.9431	0.0235	0.9431
CNMF [14]	0.0253	0.0283	0.9470
SSFCNN [50]	0.0290	0.0253	0.9464
HyperPNN [51]	0.0331	0.0309	0.9369
DHP-DARN [35]	<u>0.0249</u>	<u>0.0216</u>	<u>0.9539</u>
DIP-Hyperkite [36]	0.0272	0.0260	0.9474
Hyper-DSNet [34]	0.0259	0.0238	0.9508
HyperRefiner [38]	0.0257	0.0242	0.9505
Ours	**0.0227**	**0.0201**	**0.9575**

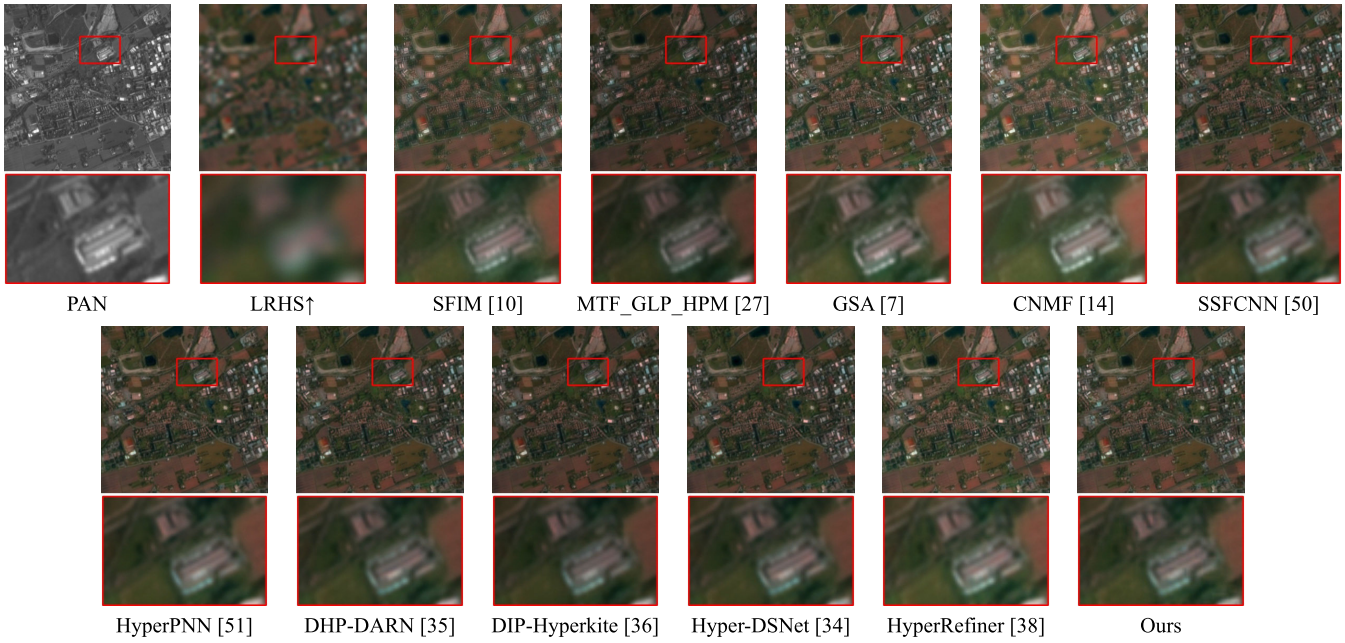


Fig. 9. Fusion results on the real FR1 dataset.

TABLE III
ABLATION STUDY RESULTS OF DFCFN AT DIFFERENT STAGES ON THE PAVIA CENTER DATASET

First stage	Second stage	PSNR (↑)	SAM (↓)	SSIM (↑)	SCC (↑)	RMSE×10^{-2} (↓)	ERGAS (↓)
✓	×	38.2866	4.7959	0.9612	0.9864	1.3539	2.6220
×	✓	37.7593	4.9849	0.9582	0.9857	1.4330	2.7411
✓	✓	**38.5797**	**4.7224**	**0.9620**	**0.9870**	**1.3112**	**2.5485**

TABLE IV
ABLATION STUDY OF DIFFERENT METHODS ON THE PAVIA CENTER DATASET

Methods	PSNR (↑)	SAM (↓)	SSIM (↑)	SCC (↑)	RMSE×10^{-2} (↓)	ERGAS (↓)	Parameters
A	36.3986	5.5638	0.9440	0.9744	1.7034	3.1548	5.13M
B	37.9415	4.8753	0.9596	0.9856	1.3975	2.7042	4.41M
C	**38.2866**	**4.7959**	**0.9612**	**0.9864**	**1.3539**	**2.6220**	5.63M

DL-based methods (SSFCNN [50], HyperPNN [51], DHP-DARN [35], DIP-Hyperkite [36], Hyper-DSNet [34], and HyperRefiner [38]). To evaluate the performance of all methods, we used six objective indicators [35], [39]: spectral cross correlation (SCC), spectral angle mapping (SAM), structural similarity (SSIM), root-mean-square error (RMSE), Erreur Relative Globale Adimensionnelle de Synthese (ERGAS), and peak signal-to-noise ratio (PSNR) in the

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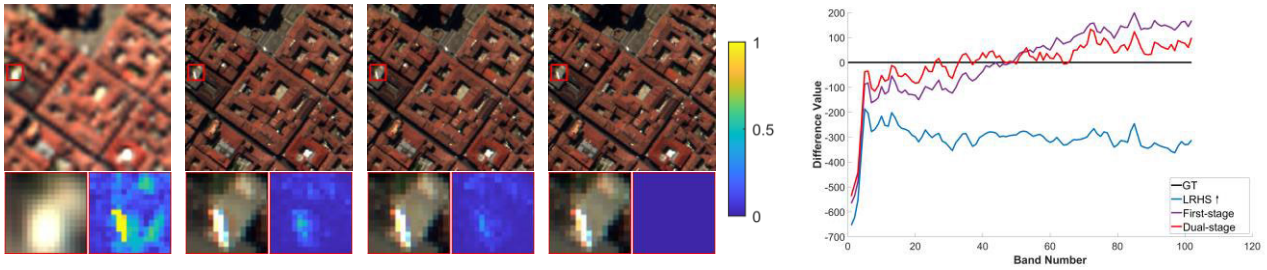


Fig. 10. Fusion results and spectrum difference curve. (a) Fusion results at different stages on the Pavia Center dataset. (b) Spectrum differences between GT and the fusion results obtained by our network at different stages.

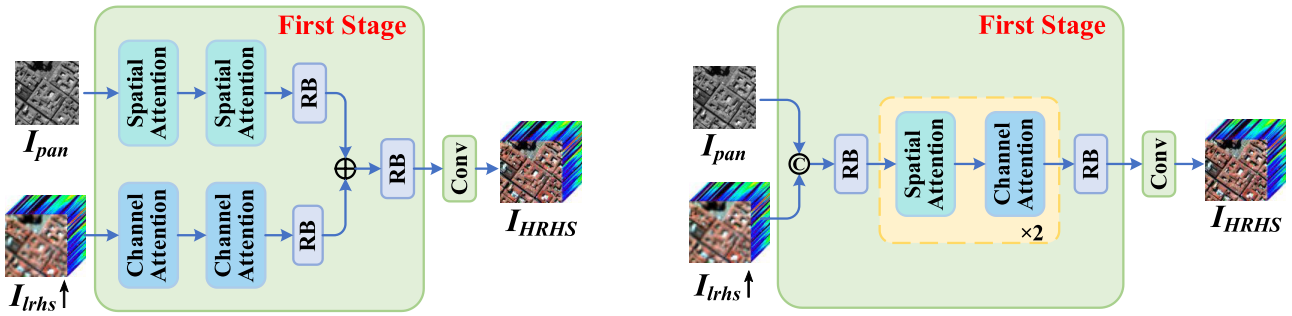


Fig. 11. Illustration of different network structures for HS pansharpening. (a) Methods A. (b) Methods B.

TABLE V
ABLATION STUDY OF FCCFN WITH DIFFERENT NUMBERS OF LFSS ON THE PAVIA CENTER DATASET

LFS	PSNR (↑)	SAM (↓)	SSIM (↑)	SCC (↑)	RMSE×10^{-2} (↓)	ERGAS (↓)	Parameters (M)
1	38.5797	4.7224	0.9620	0.9870	1.3112	2.5485	10.78
2	38.6317	4.7127	0.9621	0.9873	1.3143	2.5303	14.49

TABLE VI
ABLATION STUDY OF THE MULTIPLE SCALES STRUCTURE IN MSCFN ON THE PAVIA CENTER DATASET

Multi-level	PSNR (↑)	SAM (↓)	SSIM (↑)	SCC (↑)	RMSE×10^{-2} (↓)	ERGAS (↓)
one-level	38.4055	4.8116	0.9604	0.9862	1.3411	2.5886
two-level	38.1990	4.8351	0.9589	0.9853	1.3826	2.6391
three-level	**38.5797**	**4.7225**	**0.9621**	**0.9870**	**1.3112**	**2.5485**

TABLE VII
ABLATION STUDY OF CEB STRUCTURE ON THE PAVIA CENTER DATASET

Methods	PSNR (↑)	SAM (↓)	SSIM (↑)	SCC (↑)	RMSE×10^{-2} (↓)	ERGAS (↓)
w/o CEB	38.3755	4.8519	0.9611	0.9868	1.3368	2.5950
Ours	**38.5797**	**4.7225**	**0.9621**	**0.9870**	**1.3112**	**2.5485**

simulated experiments. In addition, to assess the performance of all methods in the real experiments, we used three metrics [34]: quality with no reference (QNR), D_λ, and D_s.

C. Experimental Results

1) Simulated Experiments: Table I presents the objective evaluation results of the fusion results obtained from all methods on the three datasets. The best results are highlighted in bold, while the second-best results are underlined. The table shows that DL-based methods exhibited better performance than traditional methods, with our proposed method outperforming the other methods on all evaluation indicators.

Figs. 5–7 show the fused images obtained from the different methods on the Pavia Center, Botswana, and Chikusei datasets,

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TABLE VIII
NUMBER OF PARAMETERS OF DL-BASED METHODS ON THE CHIKUSEI DATASET

Methods	HyperPNN [51]	DHP-DARN [35]	DIP-Hyperkite [36]	Hyper-DSNet [34]	HyperRefiner [38]	Ours
Parameters (M)	0.52	0.45	0.97	0.53	19.32	10.95
FLOPs (G)	8.86	29.06	212.64	21.89	85.00	83.43
PSNR	40.6471	41.0419	41.6503	41.6958	42.9958	**43.4760**

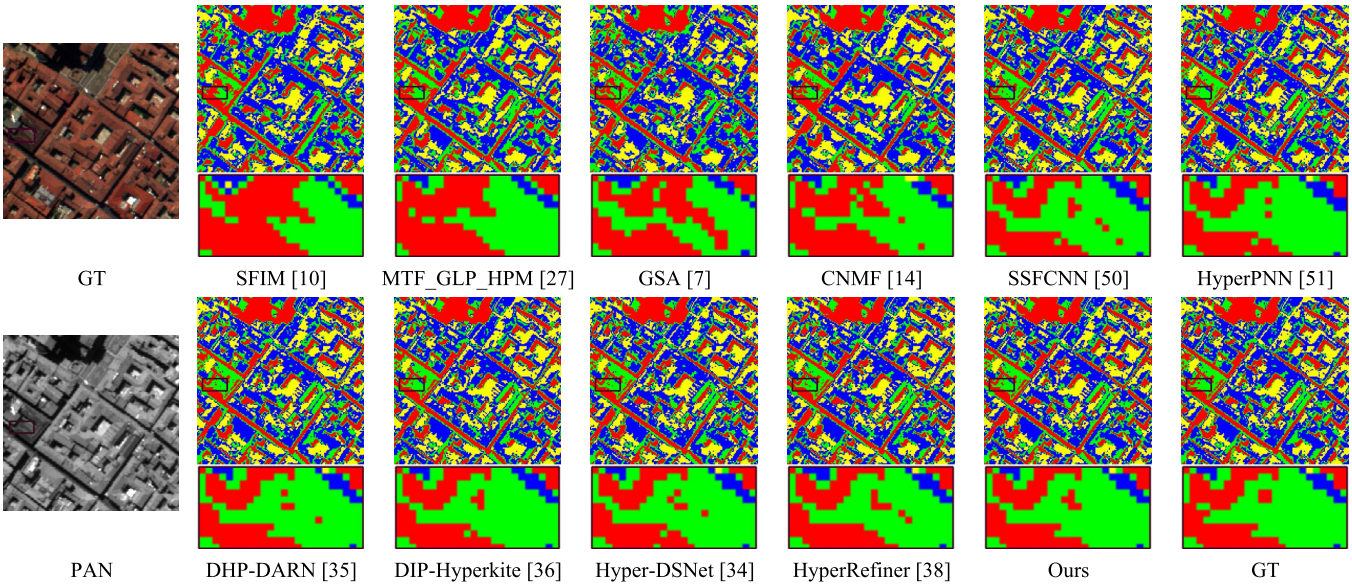


Fig. 12. Classification results of fused images from comparison methods on the Pavia Center dataset.

respectively. To observe the differences between the fusion results more clearly, the mean absolute error (MAE) maps between the fusion results and GTs were calculated and displayed, with the local areas enlarged and displayed below the results. These figures show that the results of traditional methods exhibited more severe spatial and spectral distortions compared with DL-based methods, with the corresponding MAE maps containing more residual information. The results of DL-based methods show better visual results. Our own method's results were visually closest to the GT images, and the corresponding MAE maps contained the least residual information compared with the results of other DL-based methods. This also indicates that our results had better spatial and spectral fidelity and that our proposed DFCFN outperformed other DL-based methods in the comparison.

To further assess the spectral preservation capability of the DL-based HS pansharpening methods, Fig. 8 presents the spectral difference value curves of four randomly selected pixels in three images from simulated datasets [Pavia Center (89, 130), Botswana (60, 10), Chikusei (87, 108), and Chikusei (171, 215)]. As shown in Fig. 8, our proposed method exhibited lower spectral difference values, demonstrating its excellent spectral preservation capability.

2) Real Experiments: The qualitative and quantitative results of the real experiments are shown in Fig. 9 and Table II, respectively. From Fig. 9, it can be seen that our result has sharper edges and more texture information that is closer to the PAN image. The results of other methods all have slight detail blurring issues. Table II also shows that our proposed method achieved the best results for all nonreference evaluation metrics. Therefore, it is demonstrated that our method achieves good performance on the real FR1 dataset.

D. Ablation Studies

We also performed several ablation experiments to demonstrate the value of the proposed components working together: the dual-stage structure, FCCFN, MSCFN, and CE-Transformer.

1) Effectiveness of Dual-Stage Structure: To validate the effectiveness of the network structure, we conducted ablation experiments with the results shown in Table III. The network containing dual stages achieved optimal performance. In addition, to further verify the performance of different structures, the fusion results obtained by different structures and the spectral difference value curves at a random position are shown in Fig. 10. Fig. 10(a) shows that compared with the upsampled LRHS image and the first-stage result, the fusion result obtained by the dual-stage network is closer to the GT image, and its corresponding MAE contained less residual information. Fig. 10(b) shows the spectral difference value curves at positions (89, 130), demonstrating that the result from the dual-stage network has better spectral fidelity.

2) Effectiveness of FCCFN: To verify the effectiveness of our LFS, we modified the structure of FCCFN to the structures shown in Fig. 11(a) and (b), denoted as methods A and B.

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Methods A and B include channel and spatial attention from the structures of CBAM [41]. Our first-stage network using LFS alone is represented as method C. The results are shown in Table IV. Compared with the other two structures, FCCFN using LFS achieved better performance with less increase in parameters and computational costs.

We also conducted ablation experiments on the number of LFS units. Table V presents the experimental results using different numbers of LFS, showing that the performance of FCCFN with more than one LFS is slightly improved but at the cost of many more parameters. Therefore, to balance the performance and computational complexity of the model, FCCFN's use of an LFS with two lattice units and attention interaction is the best combination.

3) Effectiveness of MSCFN: We analyzed the results obtained by MSCFNs with one-level DWT, two-level DWTs, and three-level DWTs, as shown in Table VI. The results show that the three-level DWTs in the MSCFN achieved the best performance. In MSCFNs with one- and two-level DWTs, the upsampled LRHS image is used as guidance for feature correction and supplementation. Due to the spectral distortion introduced by the upsampling operation, the indicators obtained from these two structures show a significant decrease. This also indicates that LRHS images with the original size can better correct spectral features of the initial fusion results, ensuring good spectral fidelity in the fusion results.

4) Effectiveness of CE-Transformer: We validated the effectiveness of the CEB in the CE-Transformer, with results presented in Table VII. The experimental results indicate that the CE-Transformer with CEB significantly improved the performance of the network. This also indicates that the introduction of CEB helped to utilize the spectral information of the LRHS image to correct the features of the initial fusion results.

5) Model Complexity: To evaluate the performance of the proposed network, we compared the parameters and computational complexity of our model with those of five DL-based models on the Chikusei dataset, as shown in Table VIII. Compared with HyperRefiner, we achieved better results with fewer parameters, demonstrating the performance of our proposed approach.

V. Application

To further demonstrate the effectiveness of the proposed method, we conducted downstream ground object classification experiments on the fused results. We used the k-means algorithm in the software ENVI on satellite images to evaluate the fusion results of different methods. The number of classification categories was set to 4, and the maximum number of iterations was set to 5. The classification results are shown in Fig. 12. The figure shows that the classification results from our fused images were the closest to that of the GT image, demonstrating the strength of our method.

VI. Conclusion

In this article, we propose a new two-stage fusion network called DFCFN for HS pansharpening. In the first stage, an FCCFN is constructed based on the designed LFS to achieve the initial fusion of PAN and LRHS images by correcting and supplementing the two modal features. In the second stage, MSCFN is designed by employing multilevel 2D-DWTs to obtain the final fusion result. In MSCFN, we construct CE-Transformer based on CE to correct the spectral features of the initial fusion results guided by LRHS image features. Experiments conducted on three widely used HS datasets demonstrate that our DFCFN outperforms other SOTA methods in both quantitative and qualitative evaluations. Although our proposed method effectively mitigates spectral and spatial distortions, we did not fully consider the spatial correlation between PAN and LRHS images at different scales, which may lead to the loss of texture details. In future work, we will improve the performance of the model by establishing spatial correlation at different scales between PAN and LRHS image features.

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